

Dec 15, 2025 (Due: 08:00 Dec 22, 2025)

1. Derive the following Lanczos bidiagonalization processes:

(1) The Golub–Kahan upper bidiagonalization

$$AV_k = U_k B_k, \quad A^* U_k = V_k B_k^* + [\text{rank } 1].$$

(2) The Paige–Saunders lower bidiagonalization

$$A^* U_k = V_k \tilde{B}_k, \quad AV_k = U_k \tilde{B}_k^* + [\text{rank } 1].$$

2. Let A and B be two $n \times n$ large sparse Hermitian positive definite matrices. Design an algorithm to compute a few largest eigenvalues of AB .

3. Given a black-box function that performs $x \mapsto Ax$, where A is a large-scale matrix. Suppose that you already know that $\text{rank}(A) \leq 50$. Design an algorithm to find $\text{rank}(A)$.

4. Suppose that there is a black-box function that computes $(A - \mu I)^{-1}x$, where A , μ , and x are the inputs. Describe how this black-box function can be used to solve the correction equation in the Jacobi–Davidson algorithm.

5. Implement the Lanczos algorithm for computing the matrix functional $v^* f(A) v$, where A is Hermitian. Test it with some matrices of size several thousands and some sufficiently smooth functions. Plot the convergence history.

6. (optional) Implement the LSQR algorithm. Test it with some examples and visualize the convergence history.

7. (optional) Implement the Lanczos algorithm for computing a few largest singular values and the corresponding singular vectors. Use truncated SVD to compress some grayscale images. If you only have colored images, you can convert them to grayscale using

$$\text{gray} = \alpha \cdot \text{red} + \beta \cdot \text{green} + \gamma \cdot \text{blue},$$

where $(\alpha, \beta, \gamma) = (0.299, 0.587, 0.114)$.

8. (optional) Implement the steepest descent algorithm for computing a few largest singular values and the corresponding singular vectors.